Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/28/2025
PI Name:	Robinson, Stephen K. Ph.D.		
Project Title:	A Haptic Sensory Supplement to Optimize In-Flight Adaptive Training for Human Control of Spacecraft Robotic Arms		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	95616-5270	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 HERO 80JSC019N0001-HFBP, OMNIBUS3 Crew Health: Human Factors and Behavioral Performance-Appendix E; Omnibus3-Appendix F
Start Date:	06/02/2021	**End Date:**	09/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per A. Beitman/JSC (Ed., 8/14/25). NOTE: End date changed to 06/01/2027 per A. Beitman/JSC (Ed., 8/24/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Joiner, Wilsaan Ph.D. (University of California, Davis) Schofield, Jonathon Ph.D. (University of California, Davis)		
Grant/Contract No.:	80NSSC21K1373		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Human-guided robotic arm operations will be essential for long-duration human spaceflight including LEO, cis-Lunar and Martian orbits, as well as for surface support of long-duration habitats. During these missions, astronauts will be tasked with mission-critical and demanding skill-based robotic operations long after their performance peak during pre-flight training. Therefore, they will need to re-acquire proficiency while also adapting to the differences between Earth training and mission operations. Challenges arise with long-term isolation and the accompanying somatosensory alterations during spaceflight. Thus, the astronaut requires on-board training that not only targets skill re-acquisition but is also serves as a countermeasure to neuroplastic changes and the new sensorimotor-control environment of microgravity. Finally, human-guided robotic operations involve a potentially overwhelming array of visual feedback elements that must be monitored to acquire and maintain a mental model of the robotic arm, vehicle surroundings, and near-term trajectory projections. Sensory overload is never far from reality and when risking structural collision or injuring an arm-mounted EVA astronaut, and even brief dropouts in visual scan, mental-model updates of trajectory prediction, or situation awareness can rapidly lead to mission-critical safety impacts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are addressing the NRA's proposal Topic Focus #1 by developing a class of in-flight and planetary-surface robotics adaptive training to achieve: - Adaptive acquisition of knowledge during long-duration missions (just-in-time training) - Consolidation of motor skills (transfer of training from on-board simulators to mission tasks) - Retention of motor skills (refresher training and retention assessment) - Meaningful and motivating training regimens (mission-relevant on-board training that is both appealing to the crewmembers and worth the expenditure of valuable crew time) Although robotic on-board training is currently available on ISS via the ROBoT SSRMS simulator, there is no existing spaceflight robotic training technology that is aware of and adaptive to the skill state and learning rate of individual astronauts. Thus, we propose to attenuate HRP risks by addressing somatosensory changes during long-duration missions through augmented sensory feedback to astronauts undergoing onboard just-in-time or refresher training, using experimentally-validated methods that adapt to the assessed training needs of the crew. If our research aims are achieved, we can deliver to NASA a training protocol for deep-space robotics operations that is adaptive to the current skill-state of the astronaut. By delivering a fully-trained and confident robotics operator to the surface of Mars, NASA's mission objectives and safety requirements can be met. In addition, the proposed research as an element of the MATRIKS VNSCOR is expected to lead to improvements and clarifications in NASA STD 3001a, Vol. 2, Rev.B, Part 10.6.1: "Automated and Robotic System Design".
Task Progress:	Summary of Accomplishments in Year 4: • 2025 NASA Human Research Program (HRP) Investigators' Workshop: o Presentation: "Accessing Robotic Operation Performance and Skill Transfer across Diverse Robotic Systems" (Weiwei Zhou) o Poster: "A Wearable Haptic Sleeve Interface for Spaceflight Robotic Arm Operations" (Kihun Hong) • Our research team was selected for two augmentation grants at the 2025 NASA HRP Investigators Workshop: • Designed, fabricated, and tested extensively a family of haptic sleeves with integrated vibration actuators • Extensive investigation of vibration parameters and patterns to be used for coded commanding to the arm operator through haptic signaling on the right arm • Developed reconfigurable "obstacle course" environments for testing the ability of subjects to avoid collisions while manually commanding the robot arm with haptic signaling for impending collision warning • Developed HERA-sized robot arm and obstacle course • Developed dynamic proximity assessment software that constantly computes distances between the robot arm and its surroundings as it moves under manual command • Developed and tested two different 6-DOF control methods for commanding robot arm motion • Completed a 40-subject experiment for robot-arm manipulation through 3 levels of task difficulty. Data analysis in work • Developed extensive software to enable all the above capabilities
Bibliography Type:	Description: (Last Updated: 06/02/2026)
Articles in Peer-reviewed Journals	Hong K, Zhang R, Lee KG, Young PR, Joiner WM, Robinson SK, Goncu-Berk G, Schofield SJ. "A wearable haptic sleeve interface for spaceflight robotic arm operations." JACM. 2024 Oct 5;318-22. https://doi.org/10.1145/3675094.3681943 , Oct-2024